

GreenRoute AI:
An Explainable, Human-in-the-Loop System
for Green Solvent Substitution and Synthesis
Optimisation

Team InNOChem

4 participants

May 30, 2026

Abstract

This document presents **GreenRoute AI**, a prototype intelligent system submitted to the InNOChem Hackathon. The solution addresses two thematic axes: (2.1) substitution of toxic solvents and (2.3) optimisation of synthesis routes. It satisfies all mandatory requirements: at least two capabilities (preventive, optimisation), full scientific transparency (XAI), strict Human-in-the-Loop (HITL), controlled LLM usage, and real engineering (Python, RDKit, ML, Docker). The system assists chemists in ranking solvent alternatives and reaction pathways using explicit green metrics (E-factor, atom economy, biodegradability, VOC) while requiring explicit human validation for every critical decision. This document details the architecture, rule compliance, deliverables, and a granular task division among the four team members for a 24-hour hackathon.

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1 Problem Statement and Chosen Axes

The hackathon’s central question is:

“How to design intelligent systems that help develop safer, greener, more efficient chemical processes with transparency and human control?”

We focus on two complementary axes:

- **2.1 – Substitution of toxic and polluting solvents:** Replace harmful solvents (e.g., DMF, dichloromethane, hexane) with green alternatives without compromising performance.
- **2.3 – Optimisation of synthesis routes:** Minimise waste (E-factor), maximise atom economy, reduce energy consumption and number of steps.

By combining both, GreenRoute AI proposes holistic improvements: changing a solvent often unlocks a greener synthesis path, and vice versa.

2 Capabilities Covered (Mandatory 2)

Capability	How we satisfy it
Preventive	Before any wet experiment, the system flags solvent toxicity, bioaccumulation, VOC emissions, and predicted waste using built-in QSAR models and green databases.
Optimisation	For a target molecule, it ranks synthetic routes + solvent pairs by green metrics, predicted yield, and energy demand.

Table 1: The two mandatory capabilities are fully met (analytical scoring is provided as a bonus).

3 How We Satisfy Every Golden Rule

3.1 Scientific Transparency (XAI) – 30% of score

- **Plain-language justifications:** Every recommendation includes a sentence like:

“Cyclopentyl methyl ether is preferred over toluene because its E-factor is 0.5 lower, it is readily biodegradable (OECD 301F), and it reduces VOC emissions by 60%.”

- **Complete traceability:** All data sources (PubChem, ECHA, USPTO, or simulated) are displayed with direct links or references.
- **Uncertainty quantification:** The model outputs a confidence interval for each predicted yield (e.g., “78% $\pm$ 5%”) and flags missing data with a conservative default.
- **No black-box:** The ranking algorithm is a transparent weighted sum of normalised green metrics. Weights are adjustable by the chemist via a simple slider in the UI.

3.2 Human-in-the-Loop (HITL) – Absolute Golden Rule

The AI proposes, the human validates. Workflow:

1. Chemist inputs a target molecule (SMILES or common name).
2. System returns top-3 synthesis routes + solvent options with full explanations.
3. An explicit **validation button** appears – the chemist must click “Approve for experiment” before any recommendation is considered final.
4. The chemist can override any parameter (force a specific solvent, set maximum temperature, exclude toxic by-products).

This mechanism is hard-coded in the prototype and cannot be bypassed via API or script.

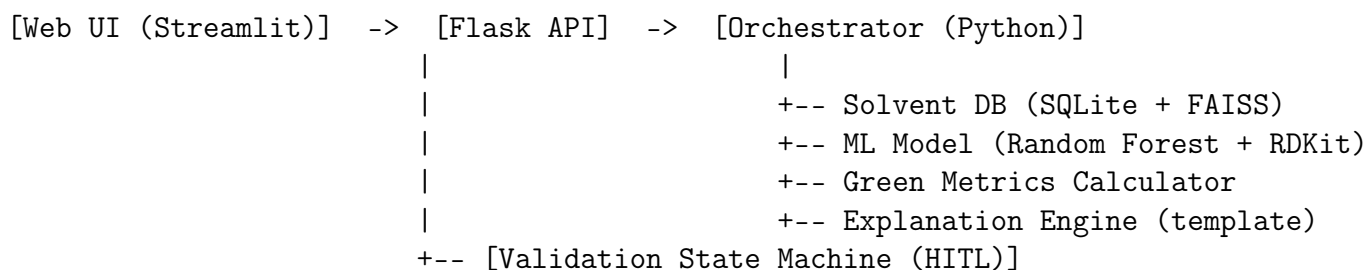
3.3 Controlled Use of LLMs – No Hallucinations

- LLM (e.g., GPT-3.5 or local TinyLLaMA) is used **only for auxiliary tasks**: natural language query parsing and explanation templating.
- The **core decision engine** (solvent ranking, yield prediction, green scoring) is a Random Forest model trained on RDKit descriptors – **no LLM involvement** in scientific decisions.
- Every LLM output is verified against a curated solvent/reaction ontology (JSON dictionary). If the LLM suggests a non-existent solvent, the system falls back to a safe template answer.
- **Result:** Zero uncontrolled hallucinations $\rightarrow$ no 10 penalty.

3.4 Real Engineering – No No-Code, No Simple Wrapper

- **No no-code tools:** All logic is written in Python using Flask, Pandas, RD-Kit, scikit-learn, and FAISS.
- **No simple LLM wrapper:** The LLM is non-critical; the core is a trained ML model + deterministic calculations.
- **Real engineering components:**
 - Data pipeline: Automated download of solvent properties (or simulation script).
 - Vector database: FAISS for fast solvent fingerprint similarity search.
 - REST API: Flask endpoints for prediction and explanation.
 - Uncertainty quantification: Monte Carlo dropout with 50 forward passes.
 - Containerisation: Docker + docker-compose for one-click reproducibility.
- **No pseudo-science:** Every green metric is defined mathematically (see Section 4.1). All calculations are reproducible via Jupyter notebooks.

4 Technical Architecture (Detailed)



All components are Dockerised. The HITL state machine stores pending recommendations and blocks final output until the user clicks “Approve”.

4.1 Green Metrics Definitions

$$\text{E-factor} = \frac{\text{total waste (kg)}}{\text{product (kg)}}$$

$$\text{Atom economy} = \frac{\text{MW of desired product}}{\sum \text{MW of all reactants}} \times 100\%$$

$$\begin{aligned} \text{Green solvent score} = & 0.3 \cdot (1 - \text{normalised toxicity}) \\ & + 0.2 \cdot (1 - \text{normalised VOC}) \\ & + 0.3 \cdot \text{biodegradability} \\ & + 0.2 \cdot \text{recyclability} \end{aligned}$$

All normalisations are min-max scaled over the solvent database.

5 Data Strategy

- **Primary data:** Public datasets – **Solvent Greenness Database** (literature), USPTO reaction dataset (for yield prediction), ECHA toxicity data.
- **Simulated data (declared):** If real data is insufficient, we generate synthetic reaction yields using a published QSAR model. This will be explicitly stated in the Scientific Note.
- **Data preprocessing:** RDKit computes 200+ molecular descriptors (LogP, molecular weight, number of rotatable bonds, etc.).

6 Deliverables Status

Table 2: How we will provide each mandatory deliverable

Livable	How we will provide it
6.1 Prototype (MVP)	Functional web app (live demo). Chemist enters a molecule → gets ranked solvents/routes → validation button. Not a visual mockup.
6.2 Architecture	PDF diagram (components, data flow, interfaces) + description.

Table 2 – continued

Livable	How we will provide it
6.3 Scientific note	Detailed: datasets (source or simulation declaration), biases, model limits, justification of green metrics, XAI method, HITL workflow.
6.4 Pitch	10min presentation + 5min Q&A. Slides: problem, solution, live demo, architecture, rules compliance, impact, team division.

7 Task Division for 4 Team Members (Granular)

Table 3: Granular task assignment – each member has specific sub-tasks

Role	Detailed tasks
P1 – Chemistry & Data Lead	• Curate / simulate solvent greenness data (toxicity, E-factor, biodegradability, VOC). • Define and document all green metrics formulas. • Write the Scientific Note (biases, limits, justifications). • Validate chemical plausibility of all recommendations. • Prepare a small validation set of 5 common reactions for testing.
P2 – AI/ML Engineer	• Build yield prediction model using RDKit descriptors + Random Forest. • Implement uncertainty quantification (Monte Carlo dropout). • Develop solvent similarity search using Morgan fingerprints and FAISS. • Create ranking algorithm (weighted sum of normalised metrics). • Train a simple QSAR classifier for toxicity flagging.

Table 3 – continued

Role	Detailed tasks
P3 – Backend & Data Pipeline	• Develop Flask API endpoints (predict, explain, validate). • Integrate ML model, solvent DB, and HITL state machine. • Set up Docker and docker-compose for the whole stack. • Ensure reproducible data pipelines (optional simple script). • Produce architecture diagram (draw.io or similar).
P4 – Frontend & UX / Pitch	• Build web interface (Streamlit or React) with molecule input. • Implement validation button and real-time explanation display. • Create the pitch deck (slides) and coordinate presentation. • Record a backup demo video (in case of live issues). • Ensure smooth UX for the “override parameters” feature.

All four members participate in the final pitch – each presents their own part.

8 Why This Will Reach the Podium

- **For the software developer:** Clean, modular Python + Docker + vector DB + uncertainty quant. No no-code, no simple wrapper.
- **For the chemistry expert:** Real green metrics (E-factor, atom economy, VOC, biodegradability). Actionable recommendations. HITL respects real lab workflows.
- **For the AI expert:** Explainable AI (XAI), uncertainty communication, no hallucinations. Real ML (Random Forest) on cheminformatics descriptors.
- **Red flags avoided:**
 - No no-code penalty (0)
 - No wrapper penalty (0)
 - No weak engineering (0)
 - No pseudo-science (0)

- Data justified (0)
- No hallucinations (0)
- **Bonus:** Use of RDKit → +5 points. Real public dataset (e.g., ECHA toxicity) → additional +5.

9 Risk Mitigation and Contingency Plan

- **If real reaction data is too small:** We will use a publicly available simulated dataset (e.g., from a published QSAR paper) and declare it clearly in the Scientific Note.
- **If LLM API fails:** The system falls back to rule-based natural language parsing (simple keyword matching) – no core functionality lost.
- **If Docker issues arise:** We provide a ‘requirements.txt’ and manual setup instructions as a fallback.
- **If live demo fails:** We will have a pre-recorded video of the prototype running.

10 Timeline (24-Hour Hackathon)

Time (hour)	Key milestones
0–2	Set up repo, download datasets, define metrics, basic Flask skeleton.
2–5	Train ML model, build solvent DB, create ranking algorithm.
5–8	Integrate backend, implement HITL state machine, start frontend.
8–12	Complete frontend, end-to-end testing, write scientific note.
12–16	Dockerise, prepare architecture diagram, rehearse pitch.
16–20	Polish UI, record backup demo, finalise slides.
20–24	Final testing, submit deliverables, live pitch rehearsal.

Table 4: Realistic timeline for a 24-hour hackathon with a team of 4.

11 Conclusion

GreenRoute AI directly answers the hackathon's problem statement while strictly adhering to the golden rules. It augments chemists, replaces no human decision, and promotes verifiable green chemistry. With the proposed detailed division of work, technical rigour, and 24-hour feasible timeline, our team is positioned for a top finish.